

The $SU(2)$ mid-plane on its small-field set: the local half of estimate (18)

Result, 2026-09-28 (Claude; written derivation, to be refereed by Fable). Take one halving of direction 1 in $1 + 2$ dimensions with gauge group $SU(2)$ and isotropic heat time $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$, as in the mid-plane note. Restrict the mid-plane fluctuations to the small-field set $\Omega_\eta = \{|\xi_e| \leq \eta \text{ for every mid-edge } e\}$ with

$$\varepsilon = t^{1/2-\delta}, \quad \eta = 2\varepsilon, \quad 0 < \delta < \frac{1}{6},$$

by the convex barrier of §1, and call $\mathcal{D}_s$ the resulting normalized defect (the definition of $\mathcal{D}$ in Hypothesis P(α) of the series/parallel note, with the fluctuation integral restricted). Uniformly in the size of the mid-plane:

- **Lemma 1 (covariant decay).** For every background mid-plane connection and every symmetric perturbation of range one and norm $\kappa < 2$ of the Gaussian Hessian, the inverse K obeys $|K^{-1}(e, e')| \leq (4 - \kappa)^{-1}((2 + \kappa)/6)^{d(e, e')}$, where d is the distance in the graph of edges sharing a face; at $\kappa = 0$, $\frac{1}{4} 3^{-d}$.
- **Lemma 2 (bridge tail).** The bridge midpoint law of a cut face with flux X satisfies $\beta(d(m, m_*) \geq r) \leq C_2 t^{-7/2} e^{-2r(r-|X|)/t}$ for $r \geq |X|/2$; at $r = \eta$ and $|X| \leq \varepsilon$ this is at most $C_2 t^{-7/2} e^{-4t^{-2\delta}}$.
- **Lemma 3 (uniform convexity).** For coarse fluxes $|X_p| \leq \varepsilon$ and $\xi \in \Omega_{c_0}$, c_0 a small absolute constant, the Hessian of the exponent in the fluctuation variables lies between $(H_A - \kappa_t)/t$ and $(H_A + \kappa_t)/t$, with $H_A = 4I + \frac{1}{2}C_A^* C_A$ (C_A the covariant curl of the background) and $\kappa_t \leq C_3(\varepsilon + |\xi|_\infty)$; on Ω_η , $\kappa_t \leq 3C_3\varepsilon$.
- **Proposition 4 (local Laplace comparison).** If the coarse data U are joined to the trivial configuration by a small-field path with controlled flux velocities (Hypothesis I of §5), then

$$|\mathcal{D}_s(U) - \mathcal{D}_s(1) - \mathcal{D}_0(U)| \leq C_4 t^{1/2-\delta} \sum_p \frac{|X_p|^2}{t},$$

with $\mathcal{D}_0$ the free-field defect and C_4 independent of the plane size. With $c_{\text{vac}} = \mathcal{D}_s(1)/N$ this is the weak form of Hypothesis P(α), $\alpha = \frac{1}{2} - \delta$, for the small-field defect, with all coupling shifts set to zero (they are of order t and fit inside the remainder).

The proof uses no cluster expansion. Brascamp–Lieb bounds the fluctuation covariance from above and Cramér–Rao from below, both in operator norm, and global gauge invariance removes the linear term; all bounds are norms of local operators, so they are uniform in N .

Two items stay open for the weak (18) on the full integral (§6): Hypothesis I for non-abelian data on large planes, and the large-field comparison $\mathcal{D} - \mathcal{D}_s$, where the naive Peierls bound fails for an explicit reason. Lemma 1 also suggests that a logarithmic size clause would replace $\min_j N_j \geq c_0/t$ for the torus windings (expected, not proved).

1. Setting

The mid-plane at a layer is a periodic two-dimensional lattice of N sites with $2N$ mid-edges (directions 2 and 3) and N mid-faces; distinct mid-planes are independent given the coarse data, so $\mathcal{D}$ is a sum over mid-planes. In the mid-vertex gauge of Proposition 1 write $m_e = m_{*e} e^{\xi_e}$, $\xi_e \in \mathfrak{su}(2) \cong \mathbb{R}^3$, with m_{*e} the geodesic midpoint between P_e and Q_e^{-1} , whose distance is the cut-face flux $|X_e|$. The exponent is

$$S(\xi; U) = \sum_e \frac{d(P_e, m_e)^2 + d(m_e, Q_e^{-1})^2 - \frac{1}{2}|X_e|^2}{t} + \sum_g \frac{d(m_{\partial g}, e)^2}{4t} + J(\xi; U),$$

the half faces at heat time $t/2$, the bridge denominator at t , the mid-faces at $2t$, and J collecting the heat-kernel and Haar amplitudes in exponential coordinates, smooth with derivatives bounded independently of t . It vanishes at $\xi = 0$ in its bridge part. Its linearization about the trivial configuration is the quadratic form of §1 of the mid-plane note, $\frac{1}{t}(2 \sum_e |\xi_e|^2 + \frac{1}{4} \sum_g |\bar{\Phi}_g + (C\xi)_g|^2)$, with Hessian H_0/t , $H_0 = 4I + \frac{1}{2}C^*C$.

The small-field restriction uses a convex barrier: $W_\eta(\xi) = \sum_e w(|\xi_e|)$ with w convex, $w = 0$ on $[0, \eta]$ and $w \rightarrow \infty$ at c_0 . The measure $\mu_U \propto e^{-S-W_\eta} d\xi$ on $\prod_e B(0, c_0)$ is then log-concave by Lemma 3, its mass outside Ω_η is bounded by Lemma 2, and $\mathcal{D}_s(U)$ is $-\log$ of its normalizing integral plus the ξ -independent normalizations of Hypothesis P(α) (bridge denominators and old-face ratios).

2. Lemma 1

Proof. Each mid-edge lies in two mid-faces and $|(C_A \xi)_g|^2 \leq 4 \sum_{e \in \partial g} |\xi_e|^2$ for every background, because the covariant curl adds four adjoint-rotated vectors. Hence $0 \leq C_A^* C_A \leq 8$ and $4 \leq H_A \leq 8$. Write $K = 6I + (K - 6I)$ with $\|K - 6I\| \leq 2 + \kappa$. The operator $K - 6I$ couples only edges that share a face, so $(K - 6I)^n(e, e') = 0$ for $n < d(e, e')$, and every matrix block is bounded by the operator norm. The Neumann series gives $|K^{-1}(e, e')| \leq \frac{1}{6} \sum_{n \geq d} (\frac{2+\kappa}{6})^n = \frac{1}{4-\kappa} (\frac{2+\kappa}{6})^d$. $\square$

3. Lemma 2

With normalized Haar measure and $\theta \in [0, 2\pi]$ the geodesic distance (the three-sphere of radius 2), the image formula of the $SU(2)$ midpoint note, Theorem 1, reads $k_t(\theta) = e^{t/8} \sqrt{8\pi} t^{-3/2} \Xi_+(\theta) / \sin(\theta/2)$. For $0 < t \leq 1$:

- *Upper bound.* $k_t(\theta) \leq C_+ t^{-5/2} e^{-\theta^2/(2t)}$. On $[0, \pi]$ use $\theta / \sin(\theta/2) \leq \pi$ for the term $w = 0$; the other images are smaller by $e^{-3\pi^2/t}$ and odd pairs vanish at $\theta = 0$. On $[\pi, 2\pi]$ pair $w = 0$ with $w = 1$: with $\psi = 2\pi - \theta$ the pair equals $e^{-(4\pi^2 + \psi^2)/(2t)} [(2\pi - \psi)e^{2\pi\psi/t} - (2\pi + \psi)e^{-2\pi\psi/t}]$, which divided by $\sin(\psi/2)$ is at most $(16\pi^2/t)e^{-\theta^2/(2t)}$ up to a constant.
- *Lower bound.* For $\theta \leq 1$ the term $w = 0$ dominates and $\theta / \sin(\theta/2) \geq 2$, so $k_t(\theta) \geq c_- t^{-3/2} e^{-\theta^2/(2t)}$.

Proof of Lemma 2. The bridge density with respect to Haar measure is $k_{t/2}(d(P, m)) k_{t/2}(d(m, Q^{-1})) / k_t(|X|)$. By the triangle inequality through m_* , $d(P, m) \geq r - |X|/2$ and $d(m, Q^{-1}) \geq r - |X|/2$ when $d(m, m_*) = r$. The bounds above give a density at most $(32C_+^2/c_-) t^{-7/2} \exp[-(2(r - |X|/2)^2 - \frac{1}{2}|X|^2)/t] = C_2 t^{-7/2} e^{-2r(r - |X|)/t}$, and Haar measure has total mass one. At $r = \eta = 2\varepsilon \geq 2|X|$ the exponent is at least $4\varepsilon^2/t = 4t^{-2\delta}$. $\square$

4. Lemma 3

Proof sketch. Each half-face term is $|\log(Ae^{\pm\xi}B)|^2/t$ with $|\log A|, |\log B| \leq \varepsilon$ in the mid-vertex frame. On $\mathfrak{su}(2)$ with $[T_a, T_b] = \epsilon_{abc}T_c$ and $|T_a| = 1$, $||x, y|| \leq |x||y|$, and the Baker–Campbell–Hausdorff remainders give $\log(Ae^\xi B) = \log(AB) + \text{Ad}(\zeta) + O(|\zeta|(|\zeta| + \varepsilon))$ with derivatives of the same order. The Hessian of $|\cdot|^2$ is $2I$, so each half-face contributes $2I/t$ up to $O(\varepsilon + |\xi|)/t$, and the pair gives the $4I/t$ of H_A/t . The mid-face term $|\log m_{\partial g}|^2/(4t)$ has Hessian $\frac{1}{2}C_A^*C_A/t$ up to $O(|\bar{\Phi}_g| + \sum_{e \in \partial g} |\xi_e|)/t$, and $|\bar{\Phi}_g| \leq \varepsilon$. The amplitude J has bounded Hessian, which is $O(t)/t$ and $t \leq \varepsilon$. All corrections are local, so their operator norm is bounded by a row sum. This gives $\kappa_t \leq C_3(\varepsilon + |\xi|_\infty)$ with an absolute C_3 , and convexity on Ω_{c_0} once $C_3(\varepsilon + c_0) < 4$. The constant C_3 is computable from the BCH remainder bounds; it is not tracked here. $\square$

5. Proposition 4

Hypothesis I (small-field path). There is a C^2 path $U(s)$, $s \in [0, 1]$, from the trivial configuration to U through coarse data with $|X_p(s)| \leq 2\varepsilon$, whose flux coordinates satisfy $\sum_p |\dot{X}_p(s)|^2 \leq C_I \sum_p |X_p|^2$ and $|\dot{X}_p(s)| \leq C_I \sum_{p' \sim p} |X_{p'}|^2$ (neighbouring plaquettes p'), where $X_p = X_p(U)$. For an abelian group the linear path $X(s) = sX$ satisfies it with $C_I = 1$.

Proof of Proposition 4. Put $g(s) = \mathcal{D}_s(U(s))$.

- (a) *No linear term.* A constant gauge transformation rotates every flux by the adjoint action and leaves Ω_η , the barrier and the normalizations invariant, so $\nabla_X \mathcal{D}_s$ at the trivial configuration is an adjoint-invariant vector in $\bigoplus_p \mathfrak{su}(2)^*$, hence zero. So $g'(0) = 0$, and likewise for $\mathcal{D}_0$.
- (b) *Second derivatives.* $\partial_X^2 \mathcal{D}_s = E_\mu[\partial_X^2 S] - \text{Cov}_\mu(\partial_X S)$ (with the normalizations, whose Hessians differ from their Gaussian versions by $O(1) = O(t)/t$), and for the Gaussian reference $\partial_X^2 \mathcal{D}_0 = \partial_X^2 S_2 - \text{Cov}_G(\partial_X S_2)$, with S_2 the linearized exponent and G the Gaussian with covariance tH_0^{-1} . Three comparisons, each as a quadratic form in the flux variation v :
 1. $\partial_X^2 S - \partial_X^2 S_2$ is a local operator with entries $O((\varepsilon + \eta)/t)$ on the support of μ , so $|\langle v, (E_\mu \partial_X^2 S - \partial_X^2 S_2)v \rangle| \leq C(\varepsilon + \eta)\|v\|^2/t$.
 2. Write $\partial_X S = \frac{1}{t}(M\xi + NX) + R$ with M, N the linear couplings and $R = O((|X| + |\xi|)^2/t)$, local. By Brascamp–Lieb and Lemma 3, $\text{Cov}_\mu(\xi) \leq E_\mu[(\text{Hess } S)^{-1}] \leq t(H_A - \kappa_t)^{-1}$, and by the Cramér–Rao inequality for location families, $\text{Cov}_\mu(\xi) \geq (E_\mu \text{Hess } S)^{-1} \geq t(H_A + \kappa_t)^{-1}$; the barrier adds only terms of the size of Lemma 2. Hence $\|\text{Cov}_\mu(\xi) - tH_A^{-1}\| \leq t\kappa_t/(4(4 - \kappa_t))$. The connection itself need not be small on a large plane, only its fluxes, so the comparison with the abelian reference is made gauge-invariantly: the flux response $M_A H_A^{-1} M_A^T$ of the covariant Gaussian differs from the abelian $M H_0^{-1} M^T$ by holonomy factors around closed paths, and by Lemma 1 a path of length L carries weight 3^{-L} while enclosing at most L^2 plaquettes of flux $\leq \varepsilon$; the sum over paths gives a difference $O(\varepsilon)$ in operator norm (sketch). So $|\langle v, (M_A \text{Cov}_\mu(\xi) M_A^T - tM H_0^{-1} M^T)v \rangle|/t^2 \leq C\varepsilon\|v\|^2/t$.
 3. By Brascamp–Lieb, $\text{Var}_\mu \langle v, R \rangle \leq \frac{t}{4 - \kappa_t} E[|\nabla_\xi \langle v, R \rangle|^2] \leq C(\varepsilon + \eta)^2\|v\|^2/t$, and the cross covariance is bounded by the geometric mean of the two variances, $C(\varepsilon + \eta)\|v\|^2/t$.

So $|\langle v, (\partial_X^2 \mathcal{D}_s - \partial_X^2 \mathcal{D}_0)v \rangle| \leq C(\varepsilon + \eta)\|v\|^2/t$ at every point of the path.

- (c) *Taylor along the path.* $g(1) - g(0) = \int_0^1 (1-s)g''(s) ds$ with $g'' = \langle \dot{X}, \partial_X^2 \mathcal{D}_s \dot{X} \rangle + \langle \nabla_X \mathcal{D}_s, \ddot{X} \rangle$, and the same identity for $\mathcal{D}_0$, which is quadratic. The first terms differ by $C(\varepsilon + \eta)C_I\|X\|^2/t$. For the second, $|\nabla_X \mathcal{D}_s| \leq C(\varepsilon + \eta)/t$ per plaquette and $|\dot{X}_p| \leq C_I \sum_{p' \sim p} |X_{p'}|^2$, which is of the same order; $\nabla \mathcal{D}_0$ obeys the same bound. With $\eta = 2\varepsilon$ and $\varepsilon = t^{1/2-\delta}$ the total is $C_4 t^{1/2-\delta} \|X\|^2/t$. No constant depends on N : every bound is the norm of a local operator or a Brascamp–Lieb estimate with local gradients. $\square$

References for the two inequalities: Brascamp and Lieb (1976) (metadata); the Cramér–Rao bound $\text{Cov} \geq I^{-1}$ with the Fisher information $I = E[\nabla S \nabla S^T] = E[\text{Hess } S]$ holds after integration by parts, which the barrier permits.

6. What stays open for the weak (18)

- (a) **Hypothesis I for non-abelian data on large planes.** For $SU(2)$ the fluxes cannot be scaled independently: link variables accumulate flux along paths, the mid-plane background carries a covariant connection, and the global holonomies of the torus constrain the plaquette data. The natural route is quasi-locality: Lemma 1 gives exponential decay of

the inverse Hessian for every background, and for uniformly convex finite-range measures the covariances decay as well (Helffer and Sjöstrand 1994, metadata). Local changes of the coarse data then change $\mathcal{D}_s$ by local amounts, and a telescoping over plaquettes replaces the global path. This is expected and not carried out here.

- (b) **The large-field comparison $\mathcal{D} - \mathcal{D}_s$.** A Peierls bound $Z(\Lambda \text{ large})/Z_s \leq e^{-\kappa(t)|\Lambda|}$ with $\kappa(t) \rightarrow \infty$ is needed for connected sets Λ of large-field edges. The naive comparison fails. Lemma 2 gains $e^{-4\varepsilon^2/t}$ per large edge at $r = \eta$. The small-field denominator bounds a face next to Λ from below only by $e^{-(|\tilde{\Phi}_g| + \sum_{e \in \partial g} |\xi_e|)^2/(4t)}$ with every $|\xi_e|$ up to η , that is by $e^{-81\varepsilon^2/(4t)}$, and each edge borders two faces: the loss exceeds the gain. Balaban's large-field methods, with gauge-invariant small-field conditions and multiscale holes, are the known way around this (Balaban 1989 I, II; for the small-field side Dimock 2013; metadata).
- (c) **Torus windings.** With Lemma 1 the torus corrections of the Gaussian part are bounded by $CN((2+\kappa)/6)^{n_*}$, $n_* = \min_j N_j$, so the weak form would need only $n_* \geq \frac{\alpha}{\log 3} \log(1/t) + C$ in place of the clause $n_* \geq c_0/t$ of the mid-plane note. This rests on (a) and is recorded as expected.

7. Consequence for STATE

Atlas cell 2: the local, small-field half of the weak estimate (18) is proved for $SU(2)$ in 1 + 2 (Proposition 4, with Lemmas 1–3), uniformly in the plane size and without a cluster expansion. Open: Hypothesis I for non-abelian data on large planes (quasi-locality), and the large-field Peierls bound, whose naive form fails for the reason in §6(b).